Klebsiella pneumoniae Sequence Type 39: An emerging cause of hospital outbreaks in Malawi

Allan Zuza*[1,2,6], Oliver Pearse[3,1], Patrick Musicha[4,1], Zoe Dyson[2,4], Kondwani Kawaza[5], Nicholas A. Feasey[6,3,1], Eva Heinz[7,3]

[1]Bacterial and Drug Resistant Infections group, Malawi Liverpool Wellcome Programme, Blantyre, Malawi [2]Department of Infection Biology, Faculty of Infectious and Tropical Diseases, London School of Hygiene & Tropical Medicine, London Wc1e 7ht, United Kingdom [3]Department of Vector Biology, Liverpool School of Tropical Medicine, Liverpool, United Kingdom [4]Parasites and Microbes Program, Wellcome Sanger Institute, Hinxton, United Kingdom [5]Kamuzu University of Health Sciences, Blantyre, Malawi [6]School of Medicine, University of St Andrews, St Andrews, United Kingdom [7]Department of Microbial and Industrial Biotechnology, University of Strathclyde, Glasgow, United Kingdom

KAMUZU UNIVERSITY OF HEALTH SCIENCES

wellcome

Introduction

Background

- There has been uneven distribution of *Klebsiella pneumoniae* STs at Queen Elizabeth Central Hospital (QECH), Blantyre over time.
- ST39 caused the majority of cases in 2017.
- 70% of these genomes were neonates.

Aim

- To investigate the genomic features causing the expansion of ST39 infections in 2017.

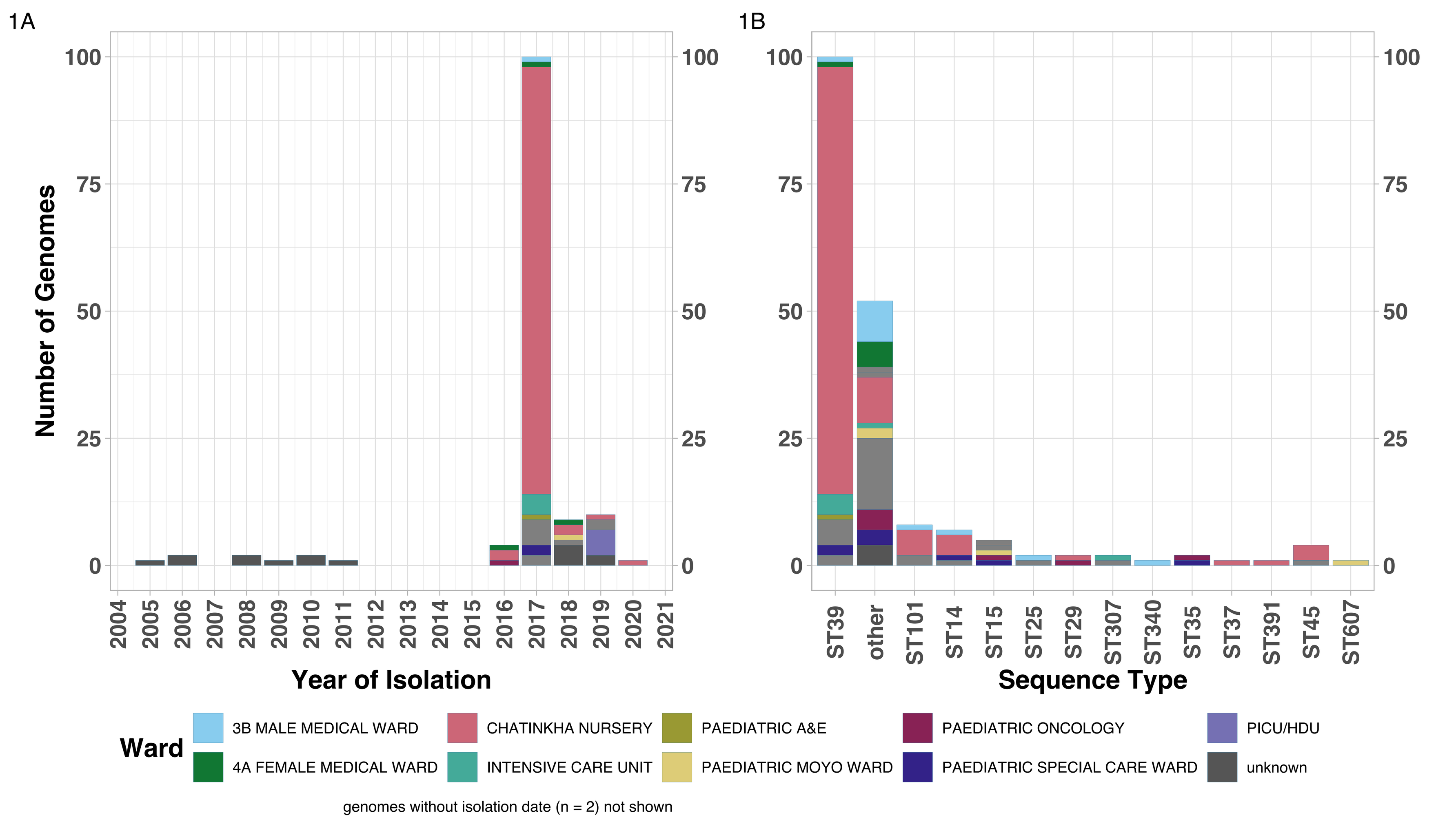
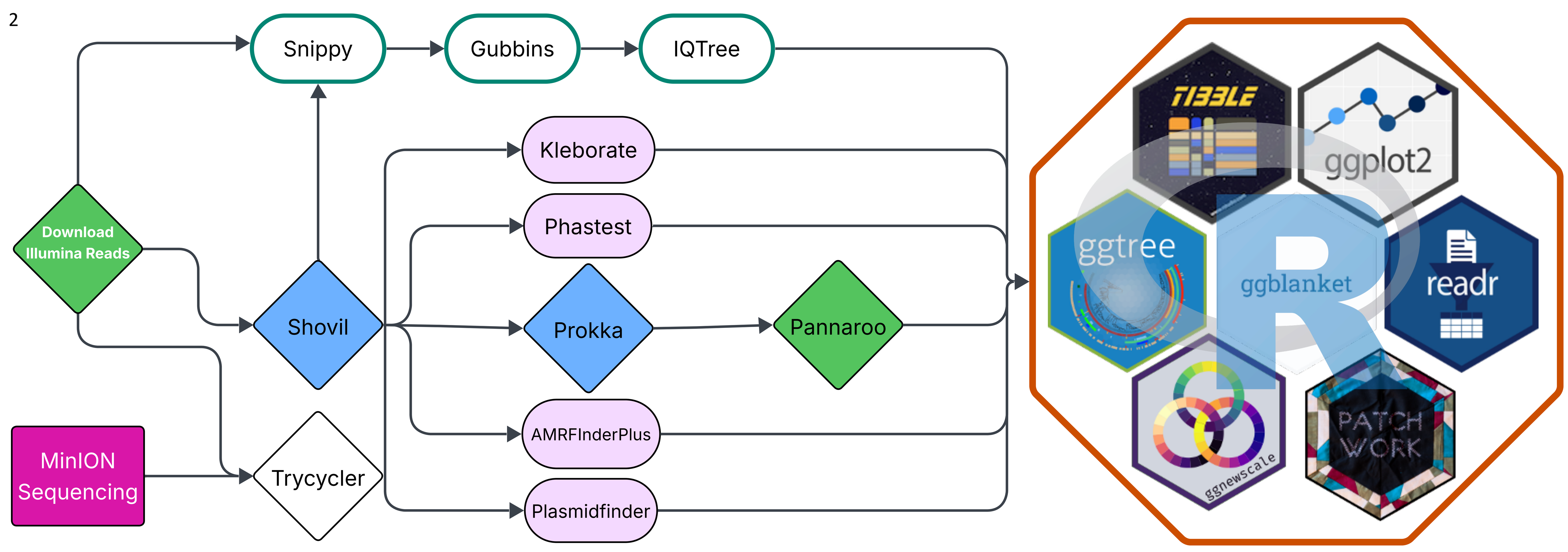


Figure 1: **A.** All available ST39 genomes from QECH showing the expansion in **2017**. **B.** All QECH Kpn from 2017 showing the number of ST39 compared to the other STs. PICU/HDU, Paediatric Intensive Care Unit/High Dependency Unit

Methods

Figure 2:

1. Downloaded **135 Illumina** genomes from ENA
2. ONT sequenced **5 representative** strains
3. Hybrid and *De novo* assembly
4. Assemblies assessed for:
   - Mobile Genetic Elements
   - virulence genes
   - K and O-types.
5. Constructed a ML phylogenetic tree
6. Data visualised in R



Results and Discussion

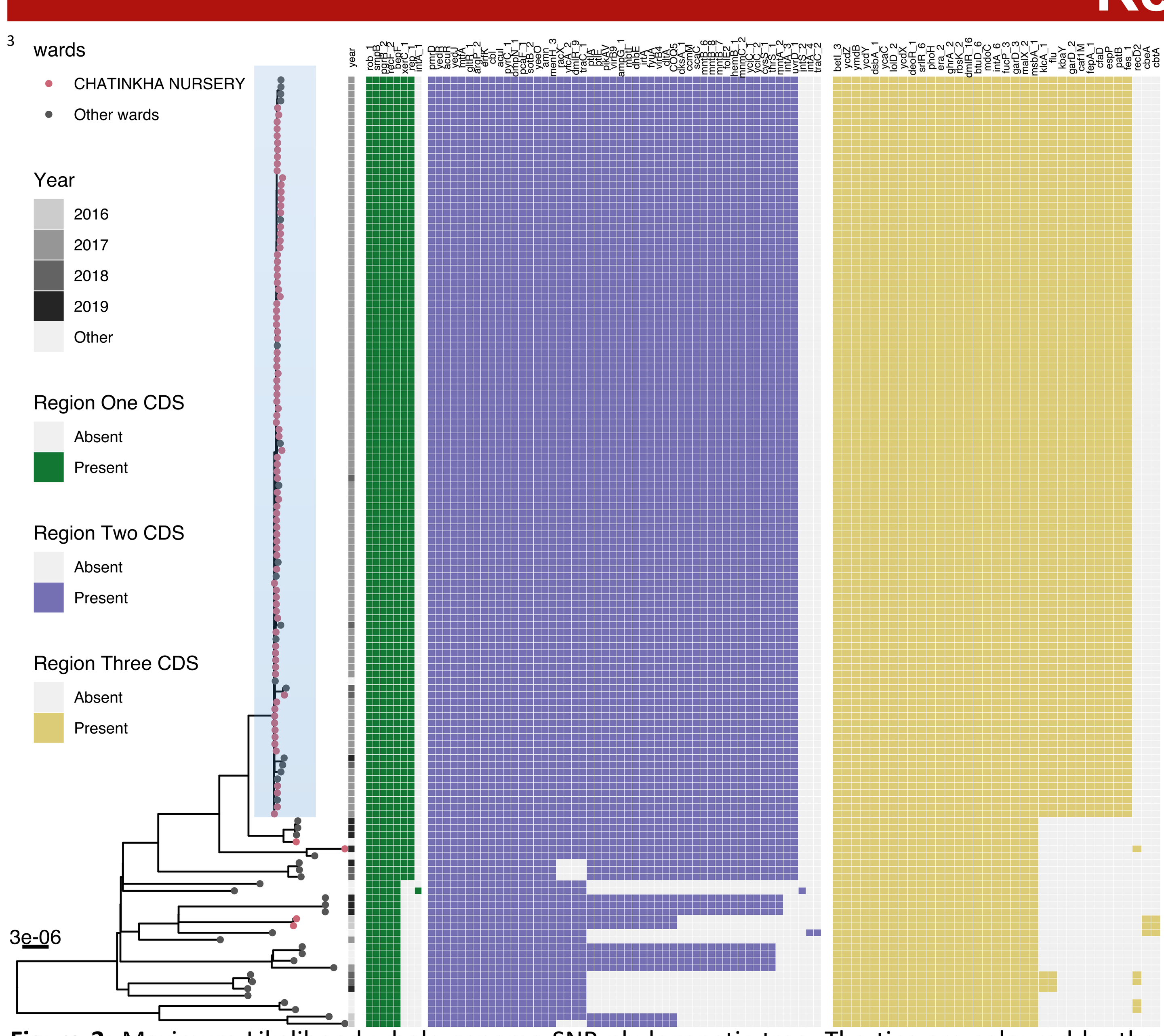


Figure 3: Maximum Likelihood whole genome SNP phylogenetic tree. The tips are coloured by the ward of isolation for the strains. The tree has the year of isolation, and three regions of variation in tRNA ser loci mapped to it on the right.

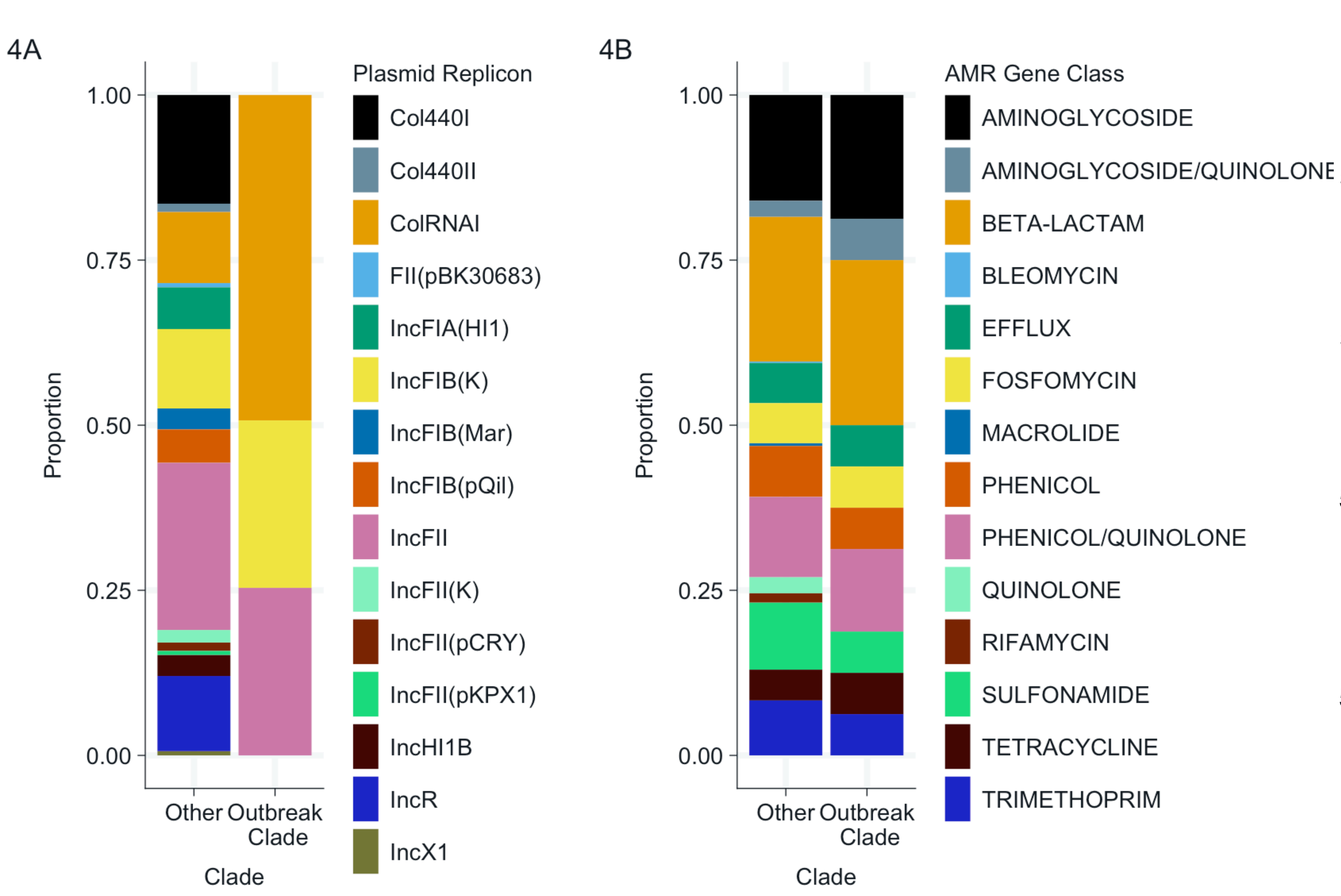
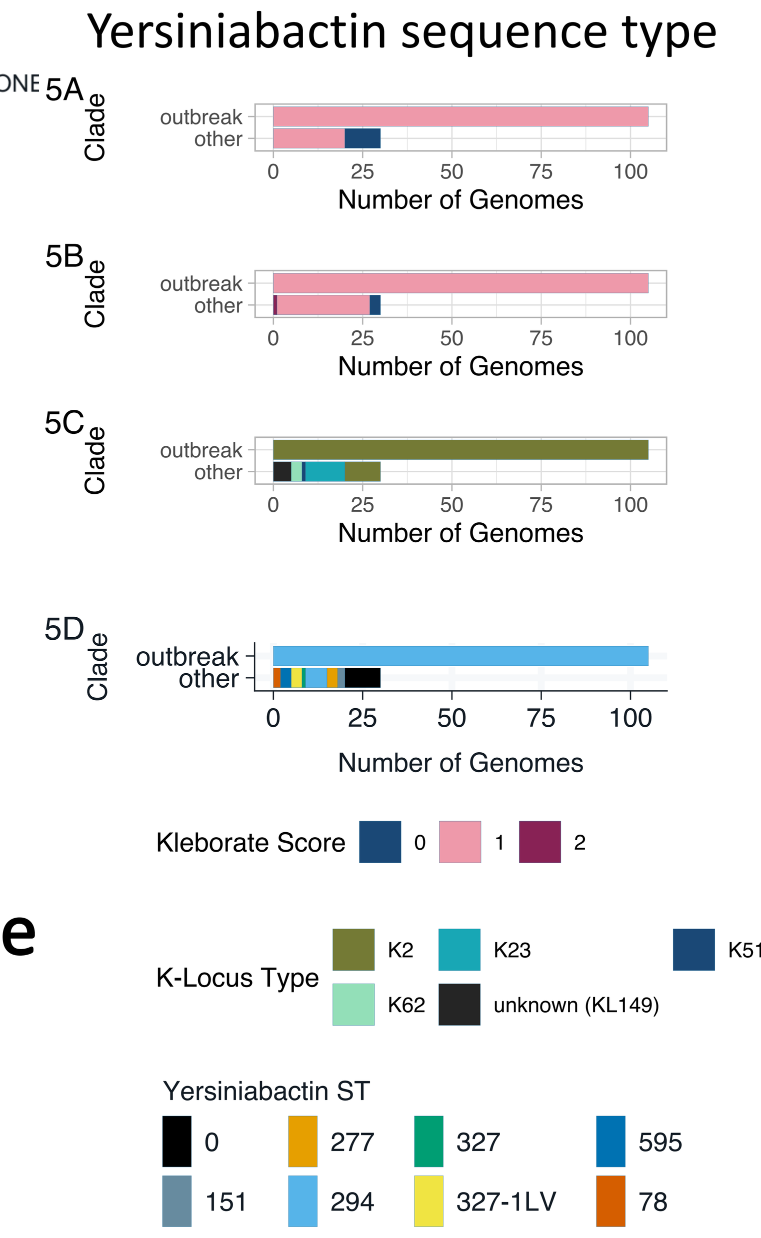


Figure 4: **A.** Plasmid replicons **B.** AMR gene class identified in the collection

Figure 5: **A.** Virulence score **B.** Resistance score **C.** K-Types **D.** Yersiniabactin sequence type



Conclusions

- Variations within tRNA-Ser loci may provide a selective advantage for K*pn* persistence in hospital environments
- ST specific reference genomes are vital for detailed analysis of genomic variations in low variant bacterial populations
- **Next steps:** **1.** Determine the function of the coding sequences in the identified regions
  2. Screen other dominant STs from Malawi for variations in the tRNA-Ser loci

References

- Heinz, E., Pearse, O., Zuza, A. *et al.* Longitudinal analysis within one hospital in sub-Saharan Africa over 20 years reveals repeated replacements of dominant clones of *Klebsiella pneumoniae* and stresses the importance to include temporal patterns for vaccine design considerations. *Genome Med* **16**, 67 (2024). https://doi.org/10.1186/s13073-024-01342-3

Acknowledgements

- Malawi Liverpool Wellcome Programme core informatics team for the compute infrastructure
- NGS Academy of the Africa CDC Pathogen Genomics Initiative for the Travel grant to this conference

LONDON SCHOOL of HYGIENE &TROPICAL MEDICINE

AfricaCDC Centres for Disease Control and Prevention

Scan me

CAMO-Net

LSTM LIVERPOOL SCHOOL OF TROPICAL MEDICINE